R20

Code No.: 20CSC22

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. V Sem (Main) Examination Dec 2022 / Jan 2023

Software Engineering

(Common to CSE, CSE-AI&ML)

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		What are the drawbacks of rapid application development model?	(3)	1	1
2		Identify the non-functional requirement-issues that are considered for any given problem description?	(3)	2	1
3		What is the role of effort estimation in a project, and why is it important to do this estimation early?	(3)	3	3
4		What is the motivation for applying the basic design principles? List the same.	(3)	4	1
5		Explain the difference between an error and a defect.	(3)	5	2

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	Define Software Engineering. Why real projects rarely use waterfall model? Explain.	(5)	1	3
	(b)	Explain why spiral model is more useful than prototype model in development of software product.	(4)	1	3
		(OR)			
7	(a)	What is the essence of software practice? Discuss the general principles of software practice.	(5)	1	3
	(b)	Compare various specialized Process Models.	(4)	1	1
8	(a)	The basic goal of the requirements activity is to get an SRS that has some desirable properties. What are the desirable characteristics of such an SRS?	(5)	2	2
	(b)	What is a spike solution in XP? Describe the XP concepts of refactoring and pair programming in your own words.	(4)	2	3
		(OR)			
9	(a)	Explain why the rapid delivery and deployment of new systems is often more important to businesses than the detailed functionality of these systems.	(5)	2	2
	(b)	Consider a conference management site which allows authors to submit papers, program chairs to assign reviewers and do the final paper selection based on reviews, and reviewers to enter the review. List the major functional aspects (use cases) for this system, along with the goals, preconditions, and exception scenarios.	(4)	2	2

10	(a)	Explain why the process of project planning is iterative and why a plan must be continually reviewed during a software project.	(5)	3	2
	(b)	Suppose you make a detailed schedule for your project whose effort and schedule estimates for various milestones have been done in the project plan. How will you check if the detailed schedule is consistent with the overall plan? What will you do if it is not?	(4)	3	
		(OR)			
11	(a)	Why data flow oriented modeling is widely used in requirement analysis notation? Explain with a suitable example to construct a data flow diagram.	(5)	3	2
	(b)	For a CAD application, 3D geometric analysis (3DGA) function is 6800LOC, user interface and control facilities is 2300 LOC, 2DGA is 5300LOC, database management is 3350LOC, computer graphics display facilities is 4950LOC, peripheral control function is 2100LOC, design analysis modules is 8400LOC, historical data review indicates the organizational average productivity for systems of this type is 620LOC/pm and burdened labour rate of Rs 8000per month. By using LOC based approach to compute: the cost per LOC, estimated effort person months and total estimated project cost.	(4)	3	3
12	(a)	How are the concepts of coupling and software portability related? Provide examples to support your discussion.	(5)	4	2
	(b)	Consider a software product of size 0.350KLOC with total minor defect are nine and major defects are five. Compute the defect density for both minor and major defects per KLOC.	(4)	4	2
		(OR)			
13	(a)	Explain the Taxonomy of Architectural Styles briefly.	(5)	4	1
	(b)	What is the goal of function oriented design? Explain with suitable example the usage of structure charts notation in function oriented design.	(4)	4	2
14	(a)	A formal technical review is effective only if everyone has prepared in advance. How do you recognize a review participant who has not prepared? What do you do if you're the review leader?	(5)	5	3
	(b)	Explain boundary value analysis with an example.	(4)	5	2
		(OR)			
15	(a)	Describe the difference between verification and validation. Do both make use of test-case design methods and testing strategies? Explain your answer.	(4)	5	2
	(b)	What is the use of flow graph notation? How do you identify number of independent paths in a flow graph? Explain with an example.	(5)	5	3

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